Cambridge (CIE) A Level Computer Science

Your notes

Encryption, Encryption Protocols & Digital Certificates

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How encryption works

What is encryption?

- Encryption is a method of **scrambling data** before being transmitted across a network
- Encryption helps to protect the contents from unauthorised access by **making data meaningless**
- While encryption is important on both **wired** and **wireless** networks, it's even more critical on wireless networks due to the data being transmitted over radio waves, making it easy to **intercept**

How is wireless data encrypted?

- Wireless networks are identified by a 'Service Set Identifier' (**SSID**) which along with a password is used to create a '**master key**'
- When devices connect to the **same wireless network** using the SSID and password they are given a copy of the master key
- The master key is used to **encrypt** data into '**cipher text**', before being transmitted
- The receiver uses the same master key to **decrypt** the cipher text back to '**plain text**'
- To guarantee the security of data, the **master key is never transmitted**. Without it, any intercepted data is rendered useless
- Wireless networks use dedicated **protocols** like **WPA2** specifically designed for **Wi-Fi** security

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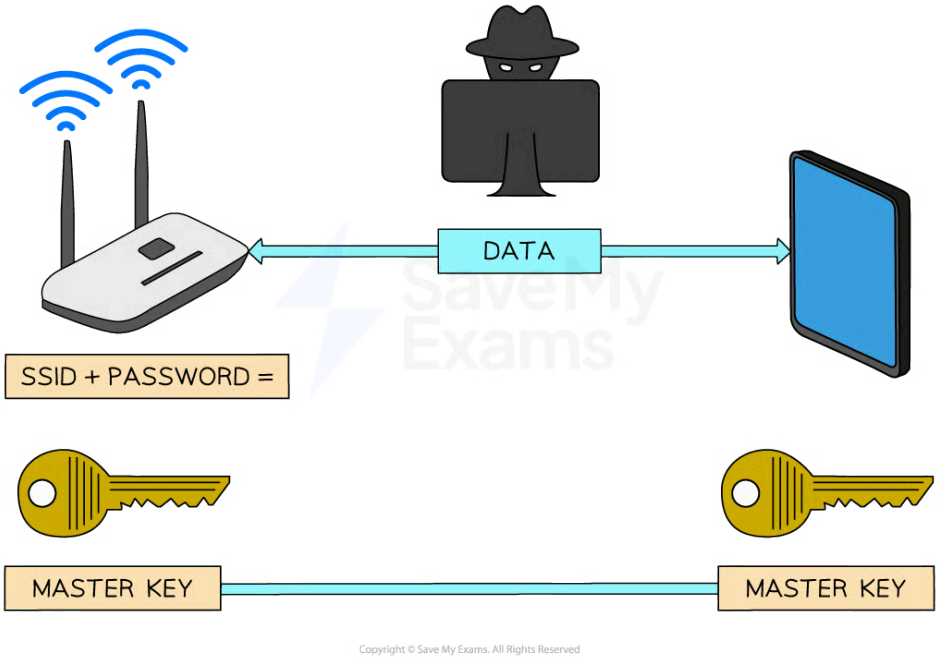


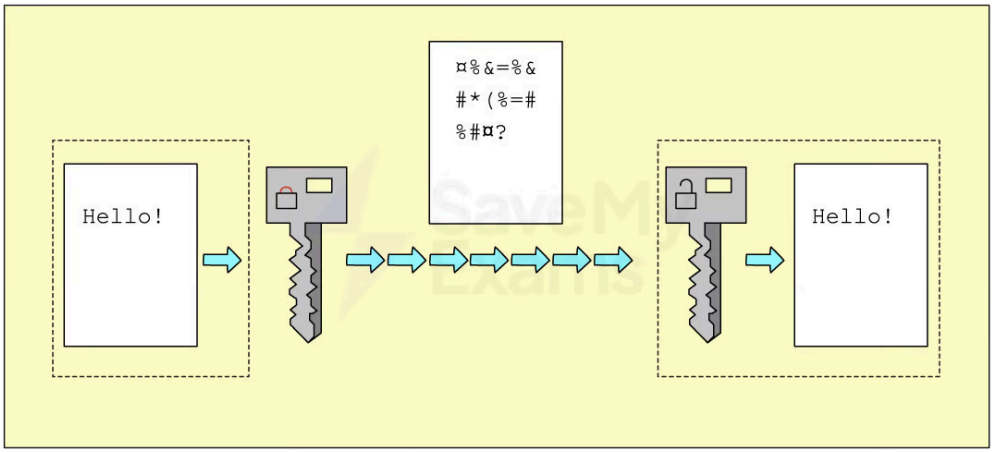
How is wired data encrypted?

- Wired networks are encrypted in a very similar way to a wireless network, using a **master key to encrypt** data and the **same key to decrypt** data
- Encryption on a wired network differs slightly as it is often left to **individual applications to decide** how encryption is used, for example **HTTPS**

Symmetric & asymmetric encryption

How does symmetric encryption work?

- The sender uses a key to **encrypt** the data before transmission
- The receiver **uses the same key** to **decrypt** the data
- It's usually **faster**, making it ideal for encrypting **large amounts of data**
- The significant downside is **the challenge of securely sharing this key** between the sender and receiver
- If a **bad actor** captures the key, they can **decrypt all messages intercepted** in transmission



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Structure of Symmetric Encryption

How does asymmetric encryption work?

- Asymmetric encryption uses **two keys**:
  - a **public key** for encryption
  - and a **private key** for decryption
- **Receivers** openly share their **public key**
- **Senders** use this public key to **encrypt** the data
- The **receiver's private key** is the **only key** that can decrypt the data and is kept locally on their side
- The public and private keys are created at the same time and are **designed to work together** in this way
- It is typically **slower than symmetric encryption**
- It is generally used for **more secure and smaller** data transactions, e.g. **passwords**, **bank details**

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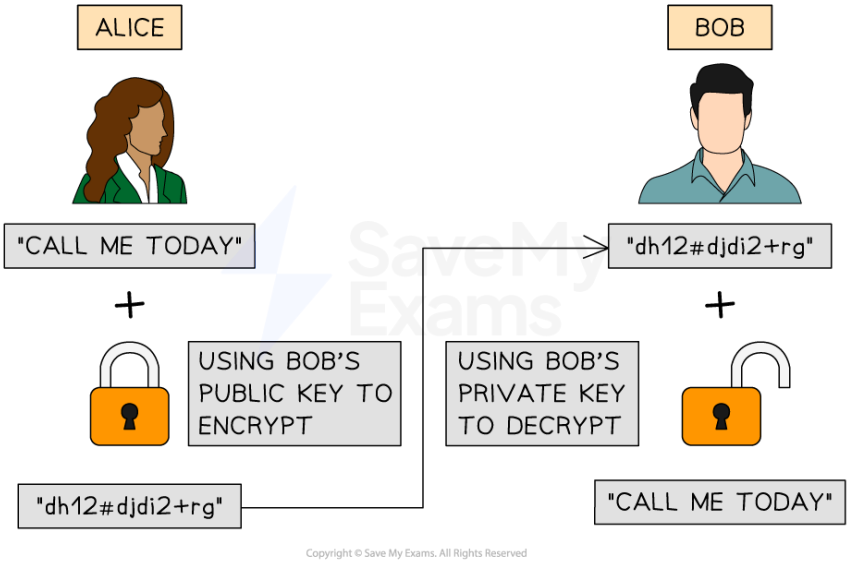

Structure of Asymmetric Encryption

Choosing an encryption type

- **Symmetric encryption is fast** but has key-sharing issues; **asymmetric is slower** but solves these issues.
- The choice should be made based on the **situation's needs**: whether **speed** or **security** is more critical.

Encryption Type	Suitable For	Reasons to choose
Symmetric	Large files, databases	▪ Fast and efficient for bulk data. ▪ The same person encrypts and decrypts, e.g. when backing up data.
Asymmetric	Confidential/secret communications	▪ Sharing highly secure data, e.g. passwords, government communications

Quantum cryptography

What is quantum cryptograhy?

- Quantum cryptography uses **quantum mechanics** to securely transmit encryption keys
- Its main goal is to enable **unbreakable communication** by detecting any attempt to intercept or tamper with the key
- The most well-known method is:

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- **Quantum Key Distribution (QKD)** – uses quantum particles (like photons) to share a secret key between two parties securely

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Benefits of quantum cryptography

Benefit	Explanation
Unbreakable key transmission	Uses quantum physics – measuring a quantum state disturbs it, so eavesdropping is detectable
Eavesdropper detection	Any interception changes the quantum state of the key, alerting the users
Perfect forward secrecy	Keys are used once and then discarded, reducing the impact of future key leaks
Stronger than classical encryption	Not based on mathematical problems like factoring large primes, so it's not vulnerable to advances in computing (e.g. quantum computers)

Drawbacks of quantum cryptography

Drawback	Explanation
Expensive and complex	Requires advanced technology, including specialised hardware like photon detectors and fibre-optic channels
Short distance limits	Works best over short ranges (limited by current fibre-optic and signal loss issues)
Slow transmission speed	QKD is typically slower than traditional key exchange methods
Still evolving	Technology is new and not yet widely available or standardised
Only secures key exchange	Quantum cryptography secures the key, but not the actual data encryption itself – traditional algorithms still needed

Examiner Tips and Tricks

Quantum cryptography isn't about encrypting the message – it's about securing the **key** used in encryption (e.g. with QKD). Always mention this to show clear understanding.

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Worked Example

Encryption is used to alter data into a form that makes it meaningless if intercepted.

Describe the purpose of asymmetric key cryptography.

Answer

- To provide better security [1 mark]
- ... by using two different keys / a public key and a private key [1 mark]
- One of the keys is used to encrypt the message [1 mark]
- ... the matching key is used to decrypt the message [1 mark]

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SSL/TLS

What is SSL?

- SSL (Secure Sockets Layer) is a **security protocol** used to **encrypt data sent over the internet**
- Helps to prevent **eavesdropping**, **tampering**, or **man-in-the-middle attacks**
- Commonly used for:
  - **Protecting websites**
  - **Online payments**
  - **Login credentials**
- Creates a **secure connection** between a web server and a browser

What is TLS?

- TLS (Transport Layer Security) is the **successor to SSL**
- It performs the **same functions** but is **more secure and efficient**
  - Modern websites use **TLS**, although people often still refer to it as "SSL"
  - When you see **HTTPS** in the browser, it means **TLS/SSL encryption is active**

How SSL/TLS works with digital certificates

1. When you visit a secure website (e.g. https://), the server sends your browser its **digital certificate**
2. The certificate contains:
   - The server's **public key**
   - Information about the website and Certificate Authority (CA)
   - A **digital signature** from the CA
3. Your browser checks that the **certificate is valid and trusted** using the CA's public key
4. If trusted, a **secure connection is established**, often using a shared secret key (symmetric encryption) created during the handshake

Summary: What a digital certificate does

Component	Purpose
Public key	Used by your browser to create secure communication

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Website identity	Shows who owns the key
Certificate Authority	Signs the certificate to confirm it's trustworthy
Browser validation	Uses the CA's public key to verify the certificate

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Examiner Tips and Tricks

- TLS has replaced SSL — but both are encryption protocols used with digital certificates
- Always mention digital certificates when explaining how websites create secure connections

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Digital certificates

What is a digital certificate?

- A digital certificate is an **electronic file** that **confirms someone's identity** and proves that a **public key** belongs to them
- It is issued by a trusted third party called a **Certificate Authority (CA)**
- A digital certificate includes:
  - The owner's **public key**
  - The owner's **identity details** (e.g. name, email, company)
  - The **expiry date** of the certificate
  - The **Certificate Authority's digital signature**

Hash function

- A hash function is a one-way algorithm that takes an input (e.g. a message) and produces a fixed-length output, called a **hash value** or **message-digest**
- **Key features:**
  - The output is always the same length, regardless of input size
  - It is **one-way** — you cannot reverse it to get the original input
  - Even a small change in input produces a completely different output
  - Commonly used in digital signatures and password storage
- Think of it as a fingerprint for data

Message-digest

- A message-digest is the **output (the hash value) produced** when a message is processed through a hash function
- **It is:**
  - A fixed-length summary of the original message
  - Unique to the message (ideally – collisions are rare)
  - Used to check whether a message has been altered
- Think of it as the unique ID or checksum of a message

How is a digital certificate acquired?

1. **Leila** wants to be able to sign documents digitally

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2. She generates a **key pair** – one private key and one public key
3. Leila sends a **Certificate Signing Request (CSR)** to a **Certificate Authority (CA)**
    1. This includes her **public key** and identity details
4. The CA **verifies Leila's identity** using documents or other checks
5. If approved, the CA **digitally signs** a certificate and sends it back to Leila
    1. This certificate contains Leila's **public key**, identity, and the CA's signature

How is a digital certificate used to produce a digital signature?

1. Leila writes a message she wants to send to **Jonas**
2. She applies a **hash function** to the message to create a **message-digest**
3. Leila then **encrypts the message-digest using her private key**
    1. This becomes her **digital signature**
4. She sends Jonas:
    - The **original message**
    - Her **digital signature**
    - Her **digital certificate**
5. Jonas:
    - Uses Leila's **public key** (from the certificate) to **verify the digital signature**
    - Uses the CA's **public key** to verify that the certificate is genuine and hasn't been forged

Summary

Step	Purpose
Certificate issued by a CA	Proves the public key belongs to the sender
Certificate includes public key	Lets others verify digital signatures
CA's digital signature on certificate	Shows it was issued by a trusted third party
Public verifies sender and message	Ensures authenticity and integrity of the message

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- The digital certificate **proves ownership** of a public key
- The digital signature **proves a message came from the claimed sender** and wasn't altered

Don't mix them up!

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